

Week 2: Forward Markets and Parity Conditions

FINA3020 at CUHK Business School

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- Forward markets and forward premiums
- Covered interest parity (CIP), executable quotes, and order-book arbitrage
- CIP funding desk; post-2008 deviations and Hong Kong interest rates
- UIP, macroeconomic expectations, and the yen carry trade
- PPP, productivity and local prices, and real exchange-rate evidence

A forward exchange contract is a bilateral agreement to buy or sell currency at an agreed-upon rate [today] to be transacted on an agreed-upon future date.

- The **value date** is the time when the exchange of currencies occurs for a deliverable forward
 - A non-deliverable forward (NDF) has net [local] cash settlement instead
- The **forward rate** $F_{t,T}$ is set at inception time t for settlement at time T
- A forward is usually a customized OTC contract between a dealer and a client
- A firm can use a forward to fix the home-currency value of an upcoming foreign cash flow

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Consider these fictional quotations for USD/HKD (how many HKD per USD, which I label compactly as $\frac{\text{HKD}}{\text{USD}}$) from Hong Kong's perspective across maturities:

Tenor / Value Date	Dealer Bid ($\frac{\text{HKD}}{\text{USD}}$)	Dealer Ask ($\frac{\text{HKD}}{\text{USD}}$)	Spread (pips)
Spot ($T = 0$)	7.8000	7.8002	2.0
6-Month Forward ($T = 180\text{d}$)	7.7970	7.7974	4.0
12-Month Forward ($T = 360\text{d}$)	7.7930	7.7936	6.0

In this specific example, forwards trade below spot ($F < S$). The 6-month bid is 7.7970 vs spot 7.8000. The USD buys fewer HKD forward, so it is a **forward discount**. Forward rates fall with maturity (7.8000 $\rightarrow$ 7.7970 $\rightarrow$ 7.7930), so there is a downward-sloping forward curve with widening spreads (2 $\rightarrow$ 4 $\rightarrow$ 6 pips).

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In interbank FX markets, dealers quote **forward points** (swap points, pips) relative to spot:

$$\text{Outright Forward Rate } F_{t,T} = S_t + \frac{\text{Forward Points}}{\text{Pip Multiplier}} \quad (\text{remember } 1 \text{ pip} = 0.0001, \text{ or } 0.01 \text{ for JPY})$$

Outright bid is *always* below outright ask ($F^{\text{bid}} < F^{\text{ask}}$), so order books never cross.

- 6-month points quote: $-30/-28$ pips
- $F^{\text{bid}} = S^{\text{bid}} - 0.0030 = 7.8000 - 0.0030 = 7.7970$
- $F^{\text{ask}} = S^{\text{ask}} - 0.0028 = 7.8002 - 0.0028 = 7.7974$
- Voice shorthand “30/28”: subtracting larger from bid preserves $F^{\text{bid}} < F^{\text{ask}}$.
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The simple annualized forward premium or discount scales the percentage difference between the forward and spot rates:

$$p_{F,\text{annualized}} = \frac{F_{t,T} - S_t}{S_t} \times \frac{\text{days per year}}{T} = \frac{\text{Forward Points}}{S_t \times \text{Multiplier}} \times \frac{\text{days per year}}{T}$$

where $S_t \equiv S_{F/H,t}$ is the direct quote (home per foreign) and T is maturity in days.

Annualized discount for our 6-month USD/HKD example ($S = 7.8000$, $F = 7.7970$, $T = 180\text{d}$):

$$p_{F,\text{USD}} = \frac{7.7970 - 7.8000}{7.8000} \times \frac{360}{180} = \frac{-0.0030}{7.8000} \times 2 = -0.0769\% \text{ per year } (-7.7 \text{ bps}).$$

Foreign-currency discount

$$F_{t,T} < S_t \implies p_F < 0$$

Forward points < 0 . Foreign currency buys fewer home currency units forward than spot.

Foreign-currency premium

$$F_{t,T} > S_t \implies p_F > 0$$

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Two Sides of the Same Trade (Close Approximation)

Because $p_{F,\text{HKD}} = -\frac{F-S}{F} \cdot \frac{360}{T} = -p_{F,\text{USD}} \cdot \frac{S}{F} \approx -p_{F,\text{USD}}$, a base currency forward discount (-0.0769%) mirrors a quote currency forward premium ($+0.0770\%$).

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The **forward curve** plots outright forward rates $F(T)$ or forward points across maturities: Overnight (ON), 1W, 1M, 2M, 3M, 6M, 12M (1Y).

Upward-Sloping (Premium)

- Quote rate $>$ Base rate ($r_q > r_b$)
- Points > 0 and grow with maturity T
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Downward-Sloping (Discount)

- Quote rate $<$ Base rate ($r_q < r_b$)
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Around 80% of international trade volume uses trade credit or other deferred payment. Example: I agree today to 1) purchase 900 German cars that will arrive at HK's port in 1 month and 2) send the payment of 36 million EUR in 3 months

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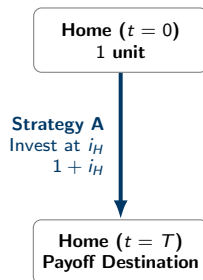
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CIP connects spot rates, forward rates, and interest rates in two currencies. Compare two investment strategies for 1 unit of domestic currency over horizon T :

Strategy A (Domestic Route)

$$\text{Payoff}_A = 1 + i_H$$



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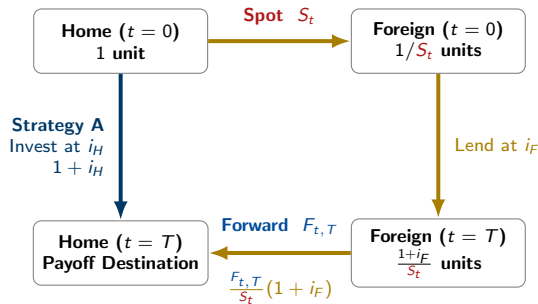
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Strategy B (Covered Foreign Route)

- Convert spot: receive $1/S_t$ foreign
- Invest abroad: earn foreign rate i_F
- Sell forward at $F_{t,T}$ to lock in:

$$\text{Payoff}_B = \frac{F_{t,T}}{S_t}(1 + i_F)$$



No Arbitrage (CIP): $\text{Payoff}_A = \text{Payoff}_B$

With matched maturities and no transaction costs, credit risk, balance-sheet costs, or capital controls, the two strategies must have the same return:

$$1 + i_H = \frac{F_{t,T}}{S_t}(1 + i_F)$$

Rearrange the equation to find the no-arbitrage forward rate:

$$F_{t,T} = S_t \cdot \frac{1 + i_H}{1 + i_F}$$

- If $i_H > i_F$, then $F_{t,T} > S_t$: the foreign currency is at a forward premium, so the higher-interest home currency is at a forward discount.
- If $i_H < i_F$, then $F_{t,T} < S_t$: the foreign currency is at a forward discount, so the lower-interest home currency is at a forward premium.

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Taking natural logarithms of the exact CIP condition $1 + i_H = \frac{F_{t,T}}{S_t}(1 + i_F)$ gives:

$$\ln(1 + i_H) = \ln\left(\frac{F_{t,T}}{S_t}\right) + \ln(1 + i_F)$$

Using first-order Taylor approximations $\ln(1 + x) \approx x$ and $\ln(F/S) \approx \frac{F - S}{S}$:

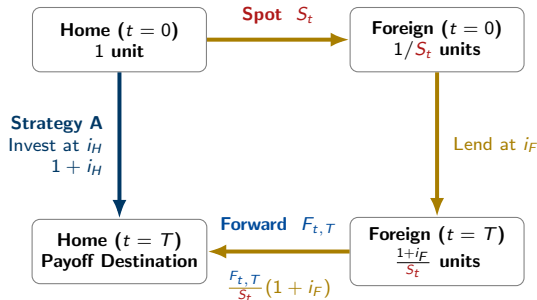
$$\frac{F_{t,T} - S_t}{S_t} \approx i_H - i_F$$

Pricing Intuition

The forward premium on the foreign currency is approximately the home interest rate minus the foreign interest rate, assuming that rates and the forward premium are close to zero and measured over the same horizon.

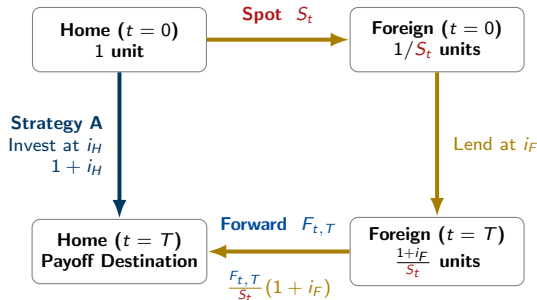
Assume $S_{\text{EUR/USD}} = 1.2000$ USD per EUR,
 $i_{\text{USD}} = 4\%$ and $i_{\text{EUR}} = 2\%$.

- ① Convert each annual rate into a 90-day holding-period return.
- ② Compare a USD deposit with a covered EUR deposit. What forward rate makes their returns equal?
- ③ Calculate the EUR forward premium over 90 days and annualize it.



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No Arbitrage (CIP): $\text{Payoff}_A = \text{Payoff}_B$

The holding-period returns are $i_{\text{USD}} = 0.04(90/360) = 0.01$ and $i_{\text{EUR}} = 0.02(90/360) = 0.005$.

$$1.01 = \frac{F}{1.2000}(1.005) \implies F = 1.2000 \frac{1.01}{1.005} = 1.205970$$

$$\text{90-day premium} = \frac{1.205970 - 1.2000}{1.2000} = 0.4975\%$$

$$\text{Annualized premium} = 0.4975\% \times 4 = 1.9900\%$$

How do interest rates shape the forward curve?

Scan the QR code and enter your details. Work and submit individually, then discuss your reasoning.

Three challenges

- ① **Dealer quotes:** Convert forward points into an outright rate and identify the premium or discount.
- ② **Corporate hedging:** Calculate the euros needed to cover a future dollar payment.
- ③ **Rate shock:** Predict how a change in interest rates shifts the forward curve.

Compare USD/HKD, EUR/USD, and USD/JPY using the **Outright Rates** and **Forward Points** views.



[Launch Forward Curve Game](#)

Exit Reflection

With fixed, unequal rates, why do longer maturities have larger absolute forward points? Submit one sentence.

What happens when CIP does not hold? Suppose $1 + i_H < \frac{F_{t,T}}{S_t}(1 + i_F)$.

An arbitrageur can use the following trade:

- ① Borrow N units of domestic currency at interest rate i_H .
- ② Convert N domestic currency to foreign currency at spot rate S_t : Receive N/S_t foreign units.
- ③ Lend N/S_t foreign units in foreign money market at interest rate i_F : Receive $\frac{N}{S_t}(1 + i_F)$ at maturity.
- ④ Sell foreign proceeds forward at rate $F_{t,T}$: Lock in domestic cash return of $\frac{N}{S_t}(1 + i_F)F_{t,T}$.
- ⑤ Repay domestic borrowing $N(1 + i_H)$.

The arbitrage profit is $N \left[\frac{F_{t,T}}{S_t}(1 + i_F) - (1 + i_H) \right]$.

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- ④ Sell foreign proceeds forward at rate $F_{t,T}$: Lock in domestic cash return of $\frac{N}{S_t}(1 + i_F)F_{t,T}$.
- ⑤ Repay domestic borrowing $N(1 + i_H)$.

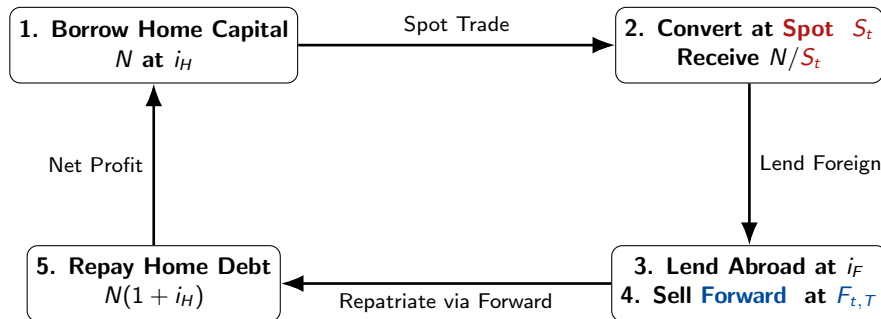
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These trades change the spot rate, forward rate, and interest rates until the profit disappears.

Market Quotes & Rates:

- **Spot Rate:** $S_{\text{GBP/USD},t} = 1.3000 \frac{\text{USD}}{\text{GBP}}$
- **1-Year Forward:** $F_{\text{GBP/USD},t,1} = 1.3200 \frac{\text{USD}}{\text{GBP}}$
- **US Interest Rate:** $i_{\text{USD}} = 2.0\%$
- **UK Interest Rate:** $i_{\text{GBP}} = 1.0\%$

Return Comparison:

$$\text{Covered GBP return} = \frac{1.3200}{1.3000}(1.01) - 1 = 2.5538\% > 2.0000\% = \text{USD return}$$

Arbitrage Result: Under the assumptions of zero cost, and simultaneous execution, borrowing \$1M USD to invest in covered GBP yields +\$5,538.46 USD.

Keep $S_{\text{GBP/USD}} = 1.3000$, $i_{\text{USD}} = 2\%$, and $i_{\text{GBP}} = 1\%$. Now let the one-year forward be $F = 1.3000$.

$$F^{\text{CIP}} = 1.3000 \frac{1.02}{1.01} = 1.312871 > F$$

Today: borrow GBP, lend USD	At maturity
Borrow GBP 1,000; sell spot for USD 1,300	USD deposit pays 1,326
Invest USD 1,300 at 2%	Buy GBP 1,010 forward for USD 1,313
Buy GBP 1,010 forward today	Repay GBP debt; retain USD 13

Why is the forward purchase GBP 1,010 rather than GBP 1,000?

Reserve: reverse CIA →

Bid-ask spreads, borrowing-lending wedges, and intermediary balance-sheet costs create a band where arbitrage is unprofitable:

- **Bid-Ask Spreads:** $S_{\text{bid}} < S_{\text{ask}}$ and $F_{\text{bid}} < F_{\text{ask}}$
- **Interest Rate Spreads:** $i^{\text{borrow}} > i^{\text{lend}}$ in both currencies

Core Intuition: Covered interest arbitrage is only executed if the return differential exceeds total transaction costs.

Takeaway: Even if you calculate a CIP deviation, you might not be able to achieve risk-free profit after considering executable quotes, funding costs, credit terms, capital charges, and balance-sheet constraints.

Bid-Ask No-Arbitrage Inequalities →

Rates below match the forward's maturity; S and F are home currency per foreign unit.

Borrow / invest	Spot execution	Forward execution
Borrow home / lend foreign	Buy foreign at S_{ask}	Sell foreign at F_{bid}
Borrow foreign / lend home	Sell foreign at S_{bid}	Buy foreign at F_{ask}

A profitable trade requires one of these inequalities:

$$(1 + i_F^{lend})F_{bid} > (1 + i_H^{borrow})S_{ask}$$

$$(1 + i_H^{lend})S_{bid} > (1 + i_F^{borrow})F_{ask}$$

Think: Can a deviation calculated with mid-quotes survive neither test? Explain which prices an arbitrageur actually receives.

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Think: Can a deviation calculated with mid-quotes survive neither test? Explain which prices an arbitrageur actually receives.

One-year AUD/HKD quotes, in HKD per AUD. Borrow and lend at $i_{\text{HKD}} = 4\%$, $i_{\text{AUD}} = 1\%$; no other costs. At the top, spot bid/ask is 5.990/6.000 and forward bid/ask is 6.200/6.210.

Level	Spot asks: buy AUD today		Forward bids: sell AUD at T	
	Price	AUD size	Price	AUD delivery size
1	6.000	800	6.200	606
2	6.010	700	6.190	909
3	6.020	1,000	6.180	1,010

- ① Compare F^{CIP} at the spot midpoint with the forward midpoint.
- ② Now use executable quotes to select the trade. Which best-quote leg limits its size?

Assume simultaneous fills, divisible quantities, and fixed books.

Walking the book algorithm

The midpoint benchmark is $5.995(1.04/1.01) = 6.17307 < 6.205$.

Borrow HKD, buy AUD spot, deposit AUD, and sell its maturity proceeds forward.

q AUD today $\longrightarrow$ $1.01q$ AUD at maturity

$$q \leq 800, \quad 1.01q \leq 606 \implies q_{\max} = \min(800, 606/1.01) = 600$$

Cash flow	Amount
HKD borrowed today: $600(6.000)$	3,600.00
HKD debt at maturity: $3,600(1.04)$	3,744.00
HKD received forward: $606(6.200)$	3,757.20
Profit at maturity	HKD 13.20

The forward leg binds. Matching both legs at 600 AUD would leave the interest unhedged.

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After the first trade, 200 AUD remain at the spot ask of 6.000
The next forward bid is 6.190 for 909 AUD delivered.

$$\text{Profit per AUD bought spot} = m(a, f) = 1.01f - 1.04a$$

$$\Delta q = \min \left(\text{remaining spot size}, \frac{\text{remaining forward delivery size}}{1.01} \right)$$

- ① At each pair of prices, compute $m(a, f)$ before taking a new tranche.
- ② Reduce spot depth by Δq and forward depth by $1.01\Delta q$.
- ③ Move down whichever book runs out. Stop when marginal profit is non-positive.

Find total spot purchases and total HKD profit. Do not round intermediate quantities.

[Order books](#)

Spot ask	Forward bid	Spot AUD	Forward AUD	HKD/AUD margin	HKD profit
6.000	6.200	600	606	0.0220	13.20
6.000	6.190	200	202	0.0119	2.38
6.010	6.190	700	707	0.0015	1.05
Total		1,500	1,515		16.63

Next pair: $1.01(6.180) - 1.04(6.020) = -0.0190$ HKD per AUD. Stop.

- Both prices worsen as we walk the books: all later margins are also negative.
- The first tranche earns the most per AUD. Check if later tranches have *positive marginal profit*.
- With uncrossed books and these common borrowing/lending rates, the reverse trade is already unprofitable at the best quotes and worsens with depth.

Reserve: forward depth →

Keep the best quotes $S_{\text{ask}} = 6.000$ and $F_{\text{bid}} = 6.200$. Let the HKD borrowing rate rise from 4% to $4\% + x$; AUD lending stays at 1%.

$$1.01(6.200) - (1.04 + x)(6.000) = 0$$

$$x^* = \frac{1.01(6.200)}{6.000} - 1.04 = 0.0036667$$

A Spread Is a Difference in Rates

The trade breaks even at an additional borrowing spread of **0.3667 percentage points**, or **36.67 basis points**.

At a one-percentage-point spread, the margin is $1.01(6.200) - 1.05(6.000) = -0.038$ HKD per AUD.

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Work on your own device, but you can discuss with classmates.

- ① Choose **borrow HKD**, **borrow USD**, or **no trade**
- ② Write the cash flow at $t = 0$ and repayment at T
- ③ Scan QR code or open [CIP Funding Desk Game](#)
- ④ Increase the spread until no-arbitrage bounds hold
- ⑤ Submit individual exit response explaining your choice

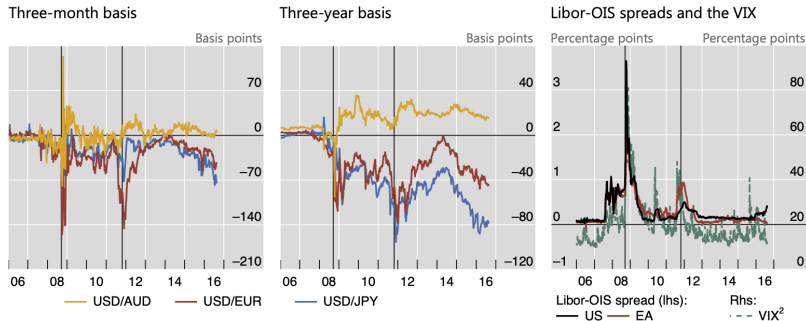
Exit Reflection

“Which executed quote and funding rate determine each no-arbitrage bound?”



[Launch CIP Desk](#)

Borrowing USD *synthetically* through FX swaps has become relatively expensive.



¹ The vertical lines indicate 15 September 2008 (Lehman Brothers file for Chapter 11 bankruptcy protection) and 26 October 2011 (euro area authorities agree on debt relief for Greece, leveraging of the European Financial Stability Facility and the recapitalisation of banks). ² Chicago Board Options Exchange S&P 500 implied volatility index; standard deviation, in percentage points per annum.

Source: Borio, McCauley, McGuire, Sushko (2016), Graph 1; historical data through 2016

Read the figure: Did the basis return to zero when the stress indicators fell? Under this figure's convention, a negative basis = CIP deviation with premium for synthetic USD borrowing.

An FX swap combines a spot exchange with a reverse exchange at a fixed forward rate. Use the foreign borrowing rate i_F . For S and F in USD per foreign unit:

$$1 + i_{\text{USD}}^{\text{synthetic}} = \frac{F}{S}(1 + i_F)$$

- Borrow foreign currency, sell it spot for USD, and buy the foreign repayment forward.
- Example: $S = 1.20$, $F = 1.236$, $i_F = 2\%$ for one year. Synthetic USD funding costs $(1.236/1.20)(1.02) - 1 = 5.06\%$.
- If direct USD borrowing costs 4.80%, synthetic funding is **26 basis points more expensive** before further execution costs.

Sign Convention

Synthetic minus direct USD cost. A positive cost premium corresponds to a negative foreign-leg basis under the preceding figure's convention.

- ① Firms and investors demand FX hedges, creating persistent pressure on forward prices.
- ② A bank taking the opposite position uses funding, credit limits, collateral, and balance-sheet capacity even when its currency exposure is hedged.
- ③ Leverage requirements and internal risk limits make that capacity costly. The bank expands only if the return covers its marginal costs.

Example: A 26 bp gross spread is unattractive if additional execution and balance-sheet costs are 30 bp in total.

Think: Why might the gap persist outside a financial crisis?

Source: Borio et al. (2016).

Under the Linked Exchange Rate System, HKMA convertibility undertakings bound USD/HKD at 7.75 and 7.85 for licensed banks.

Pressure	HKMA transaction	Adjustment in HKD money markets
Weak HKD: 7.85	Buys HKD, sells USD	Aggregate Balance contracts; upward pressure on HKD rates
Strong HKD: 7.75	Sells HKD, buys USD	Aggregate Balance expands; downward pressure on HKD rates

The Aggregate Balance is banks' clearing balances at the HKMA.

Implication: Open capital mobility and the exchange-rate commitment constrain Hong Kong's independent monetary policy. HKD rates track USD rates over time, with local liquidity affecting the spread.

Source: HKMA (2018), Linked Exchange Rate System operations. Full regime analysis: Week 3.

CIP for USD/HKD, with simple annual rates and year fraction τ :

$$\frac{F}{S} = \frac{1 + i_{\text{HKD}}\tau}{1 + i_{\text{USD}}\tau}$$

Hypothetical 90-day example: $S = 7.8000$, $r_{\text{HKD}} = 3\%$, $r_{\text{USD}} = 5\%$, and $\tau = 90/360$ for both rates.

$$F = 7.8000 \frac{1.0075}{1.0125} = 7.76148$$

- A small forward discount can offset a material *annualized* interest gap over a short horizon.
- Exact equality of rates follows from $F = S$ under frictionless CIP. The actual peg has a band and changing liquidity conditions.

Source: HKMA (2018). Hypothetical rates. No forward-rate guarantee from HKMA.

Hypothetical one-year choice: JPY deposit rate 0%, USD deposit rate 5%, and $S_t = 150$ JPY per USD. Start with JPY 150,000.

	Keep yen	Convert to dollars
Deposit today	JPY 150,000	USD 1,000
Balance in one year	JPY 150,000	USD 1,050
Yen value if $S_T = 150$	JPY 150,000	JPY 157,500
Yen value if $S_T = 135$	JPY 150,000	JPY 141,750

Break-even exchange rate: $150,000/1,050 = 142.86$ JPY per USD.

A sufficiently strong yen erases the extra interest. An unhedged investor needs a view about the future exchange rate.

Source: Illustrative rates and payoffs. Ignore fees, taxes, and credit risk. S rises when the yen depreciates.

The foreign deposit pays $(1 + i_F)S_T/S_t$ in home currency. With risk-neutral investors and comparable assets, equilibrium equalizes *expected* returns:

$$1 + i_H = \frac{\mathbb{E}_t[S_T]}{S_t}(1 + i_F), \quad \mathbb{E}_t\left[\frac{S_T - S_t}{S_t}\right] \approx i_H - i_F.$$

- In the yen example, UIP requires $\mathbb{E}_t[S_T] = 150/1.05 = 142.86$.
- The higher-interest dollar must be expected to lose value against the yen enough to offset its interest advantage.
- $\mathbb{E}_t$ uses information available *today*. Tomorrow's realized exchange rate can be very different.

In the same yen example, frictionless CIP sets the one-year forward at 142.86 JPY per USD.

Object	What determines it?
Forward $F_{t,T}$	Today's spot and matched funding rates price an exchange contract.
UIP benchmark	Risk-neutral expected-return equality implies $\mathbb{E}_t[S_T] = 142.86$.
Investor forecast	A probability distribution based on the investor's economic assumptions and information.

An investor could reasonably forecast $S_T = 146$ while the forward is 142.86. The resulting expected gain involves exchange-rate risk.

A forecast needs a **currency pair, horizon, and economic story**. Compare developments in both economies.

Evidence	Economic question	Possible currency implication
Inflation, wages, jobs	Will the central bank keep rates higher than previously expected?	A higher expected relative policy path can support the currency today.
Growth, energy, trade	Is growth attracting investment, or raising the import bill?	Capital inflows and import payments might have opposite effects.
Risk and funding	Are investors seeking safety or closing leveraged positions?	Funding currencies can rise as borrowers repay them.

Observationally equivalent but counterfactually different: high interest rates have several causes, including tight monetary policy, high expected inflation, or compensation for risk. Each has different FX implications.

Source: Policy channel: Curcuru, Federal Reserve (2017). Funding channel: BIS Bulletin 90 (2024).

Hypothetical policy announcement: A central bank raises its rate by 25 basis points.

Before the decision	New information	Likely pressure, all else equal
No hike expected	More tightening than expected	Currency strengthens
A 50 bp hike expected	Less tightening than expected	Currency weakens
25 bp fully expected	Little news in today's rate	Future guidance

Where to look: Compare releases with survey forecasts. Compare the expected policy path before and after the news using rate futures or overnight indexed swaps (OIS).

Market prices contain risk premia

Source: Curcuro, Federal Reserve (2017). The scenarios are illustrative.

US home, UK foreign: S is USD per GBP. For this one-year illustration, hold the expected future spot rate at $\mathbb{E}_t[S_T] = 1.25$.

	Before the surprise	After the surprise
US / UK interest rate	3% / 3%	5% / 3%
Spot consistent with UIP	1.2500	$1.25(1.03)/1.05 = 1.2262$
Expected future spot	1.2500	1.2500

- **On the news:** S falls. The dollar appreciates immediately.
- **After that jump:** S is expected to rise from 1.2262 to 1.25. Expected dollar depreciation offsets its higher interest rate.

Return to the yen saver: $S_t = 150$ JPY per USD, one-year rates 0% in JPY and 5% in USD.

Hypothetical one-year scenario	Probability	S_T
US rates ease gradually, Japan normalizes	50%	145
US inflation persists, relative US rates stay high	30%	155
US slowdown and rapid closing of yen-funded trades	20%	135

- ① Calculate $\mathbb{E}_t[S_T]$ and the expected yen return on the USD deposit.
- ② How large is the yen loss in the last scenario?
- ③ Which new release would make you revise the probabilities?

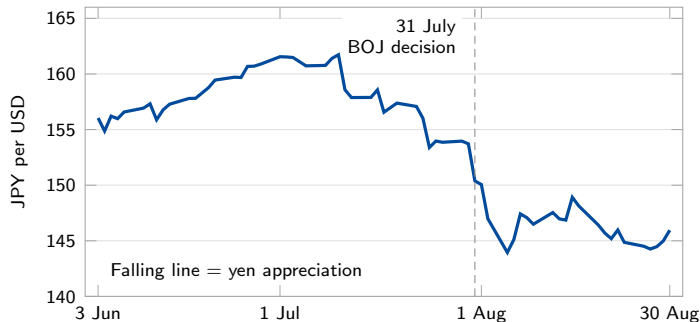
$$\mathbb{E}_t[S_T] = 0.5(145) + 0.3(155) + 0.2(135) = 146.$$

$$\mathbb{E}_t[R_{\text{USD deposit in JPY}}] = 1.05 \frac{146}{150} - 1 = \boxed{2.2\%}.$$

The JPY deposit earns 0%. The forecast implies a positive expected excess return from taking dollar risk.

$$\text{Return if } S_T = 135 : \quad 1.05 \frac{135}{150} - 1 = \boxed{-5.5\%}.$$

- The mean forecast differs from the UIP benchmark of 142.86.
- Weak US employment data might increase the probability of the last scenario. That lowers the mean forecast and the expected dollar return.
- A firm with a yen account payable may prefer certainty even when the expected dollar return is positive.



Background: Low Japanese rates made yen borrowing attractive for financing positions abroad.

Changing outlook: The BOJ raised its policy rate to around 0.25% on 31 July. Weak US data increased concern about a slowdown.

Amplification: Investors closing leveraged trades bought yen to repay funding. Yen appreciation magnified losses on remaining positions.

Why can a few days of exchange-rate losses overwhelm months of interest income?

Source: Federal Reserve H.10 via FRED, DEXJPUS, daily June–August 2024. Interpretation: BIS Bulletin 90 (2024).

The economic test: Does subsequent depreciation of high-interest currencies offset their interest advantage on average?

$$\underbrace{\frac{S_T - S_t}{S_t}}_{\text{realized home depreciation}} = \alpha + \beta \underbrace{\frac{F_{t,T} - S_t}{S_t}}_{\text{forward premium}} + \varepsilon_T.$$

- CIP, UIP, and rational expectations predict $\alpha = 0$, $\beta = 1$.
- Many historical short-horizon samples find $\beta < 0$. High-interest currencies often fail to depreciate enough, leaving positive average carry returns.
- Risk premia and systematic expectation errors are competing explanations. Estimates change across currencies, periods, and horizons.

Source: Bussière et al., The New Fama Puzzle.

Optional: calculate a Fama slope

In frictionless, competitive markets, the law of one price says an identical tradable good should have the same price when expressed in a common currency

- Necessary: same place, time, quality, and contract terms

Without trade costs:

$$P_{i,H} = S_{F/H} \cdot P_{i,F}$$

where $P_{i,H}$ is the home price of good i , $P_{i,F}$ is its foreign price, and $S_{F/H}$ is home currency per unit of foreign currency.

If $P_{i,H} < S_{F/H} \cdot P_{i,F}$, trade arbitrageurs buy good i in the home market and sell it in the foreign market until prices align.

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The same sweater (same brand, size, etc) is priced at HKD 200 in Hong Kong and USD 30 in the US. The exchange rate is 7.8 HKD per USD.

$$\text{Hong Kong purchase cost in USD} = 200/7.8 = 25.64$$

$$\text{Gross margin} = 30(7.8) - 200 = \text{HKD } 34 \text{ per sweater}$$

- ① Where should a trader buy and sell? What happens to prices as trade expands?
- ② If shipping, tariffs, and handling cost HKD 40 per sweater, is the trade profitable?
- ③ Would an identical argument work for a haircut or an apartment?

Assume a fixed conversion rate and a guaranteed sale for the calculation. Actual goods trade also involves timing, quality, and demand risk.

Absolute PPP extends the law of one price from one good to the same representative consumption basket in two countries. Absolute PPP:

$$S_{F/H}^{\text{PPP}} = \frac{P_H}{P_F}$$

where P_H is domestic price level of consumption basket and P_F is foreign price level. The nominal exchange rate equals the ratio of the two price levels.

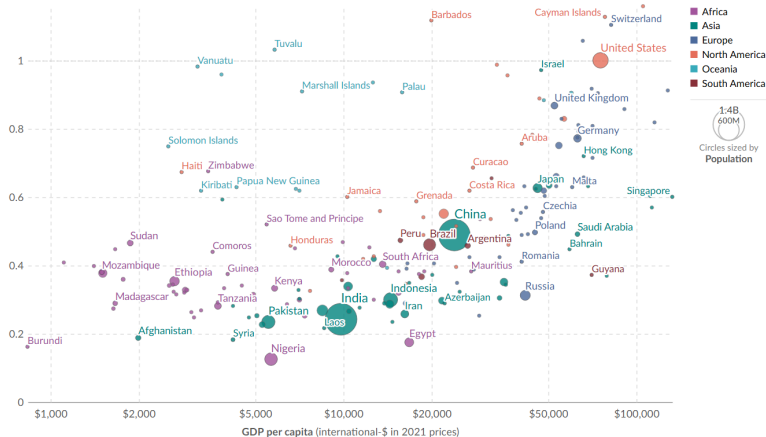
Is this realistic?

- Haircuts, housing, and many health services cannot be shipped across borders.
- Shipping costs, tariffs, and non-tariff barriers create a band within which goods arbitrage is not profitable.
- Wages and prices adjust slowly.
- Firms can set different prices in separated markets.
- National price indexes use different baskets, quality adjustments, and expenditure weights.

Richer Countries Tend to Have Higher Price Levels

[Background Context]

Price levels relative to the US



The pattern: A comparable consumption basket often costs more in richer economies after currency conversion.

The puzzle: If workers produce more efficiently, why would things cost more?

Separate the price of a traded product from the price of local labour-intensive services.

► 1990 ————— 2024

Data source: World Development Indicators - World Bank (2025); Eurostat, OECD, IMF, and World Bank (2025) - [Learn more about this data](#)

Source: Existing course figure, Our World in Data, World Bank and Eurostat inputs, displayed year 2024.

All prices and wages use a common currency. Compare the same economy before and after a factory increases productivity.

	Before	After
Tradable output per worker-hour	1	2
Tradable price	10	10
Hourly wage in both sectors	10	20
Haircuts per worker-hour	1	1
Haircut price	10	20
Cost of one tradable + one haircut	20	30
Hours of work needed to buy that basket	2	1.5

The basket price $\uparrow$ 50%, but wages double. Higher prices can coexist with improved purchasing power.

Let A_T and A_N be output per worker-hour in tradables and nontradables. With a common domestic wage w and competitive labour-only production:

$$w = P_TA_T = P_NA_N, \quad \boxed{\frac{P_N}{P_T} = \frac{A_T}{A_N}}.$$

Productivity change	Relative service price P_N/P_T
Factory productivity doubles, salon unchanged	Doubles
Factory and salon productivity both double	Unchanged
Salon productivity doubles, factory unchanged	Halves

Local wages help explain expensive haircuts and meals. Rent, taxes, and product quality also matter, so a city price comparison is not a pure test of this model.

The real exchange rate $Q_{F/H,t}$ is the price of the foreign basket relative to the home basket:

$$Q_{F/H,t} = \frac{S_{F/H,t} \cdot P_{F,t}}{P_{H,t}}$$

Interpretation:

- If $Q_{F/H,t}$ rises, the home currency depreciates in real terms. The home basket becomes cheaper relative to the foreign basket.
- If $Q_{F/H,t}$ falls, the home currency appreciates in real terms. The home basket becomes more expensive.
- Absolute PPP implies $Q_{F/H,t} = 1$.

In the factory example, the home basket price rises from 20 to 30. Suppose the foreign basket price and nominal exchange rate stay fixed.

$$Q = \frac{SP_F}{P_H}, \quad \frac{Q_{\text{after}}}{Q_{\text{before}}} = \frac{20}{30} = \frac{2}{3}.$$

- Q falls: home real appreciation. Home services become more expensive relative to foreign services.
- This can be an equilibrium response to rising tradable productivity.
- With a nominal peg, relative prices adjust more. Otherwise, the nominal exchange rate can also adjust.

Home is the US. Foreign is the UK.

A comparable basket costs USD 150 in the US and GBP 100 in the UK.

The spot quote is $S_{\text{GBP/USD}} = 1.3500$ USD per GBP.

- ① Express the UK basket price in USD, then calculate $Q = SP_{\text{UK}}/P_{\text{US}}$.
- ② Calculate the absolute-PPP exchange rate. Which currency is overvalued relative to that benchmark?
- ③ If only the nominal exchange rate adjusts to absolute PPP, what percentage change in S is required?

Use prices of comparable baskets. In reality, separately rebased national CPI indexes do not identify absolute price-level differences.

$$P_{\text{UK}} \text{ in USD} = 1.35(100) = 135, \quad Q = 135/150 = 0.90$$

The home basket is more expensive: the USD is overvalued *relative to absolute PPP*.

$$S^{\text{PPP}} = 150/100 = 1.50 \text{ USD/GBP}, \quad \frac{1.50}{1.35} - 1 = 11.11\%$$

- The GBP price in USD must rise 11.11% to reach this PPP benchmark if basket prices stay fixed.
- In the inverse quote, the USD's GBP value falls 10%. Percentages depend on the denominator.

Relative PPP focuses on changes in exchange rates and inflation differentials over time. Relative PPP:

$$\frac{S_{F/H,t+1} - S_{F/H,t}}{S_{F/H,t}} \approx \pi_{H,t+1} - \pi_{F,t+1}$$

where π_H is home inflation rate and π_F is foreign inflation rate. If home inflation is higher, the home currency should depreciate over a long horizon by approximately the inflation differential.

From $Q = SP_F/P_H$, the exact one-period change is

$$\frac{Q_1}{Q_0} = \frac{S_1}{S_0} \frac{1 + \pi_F}{1 + \pi_H}$$

Constant Q: If US inflation is 4% and UK inflation is 2%,

$$\frac{S_1}{S_0} = \frac{1.04}{1.02} = 1.019608$$

The GBP price in USD rises 1.96%, approximately the 2 percentage point inflation gap.

Changing Q: Suppose instead S rises 5%, UK inflation is 5%, and US inflation is 2%:

$$\frac{Q_1}{Q_0} = 1.05 \frac{1.05}{1.02} = 1.080882$$

Real home depreciation is 8.09%, approximately $5 + 5 - 2 = 8\%$. Relative PPP would require Q to stay constant, which it does not in this second case.

Reserve: peg and inflation →

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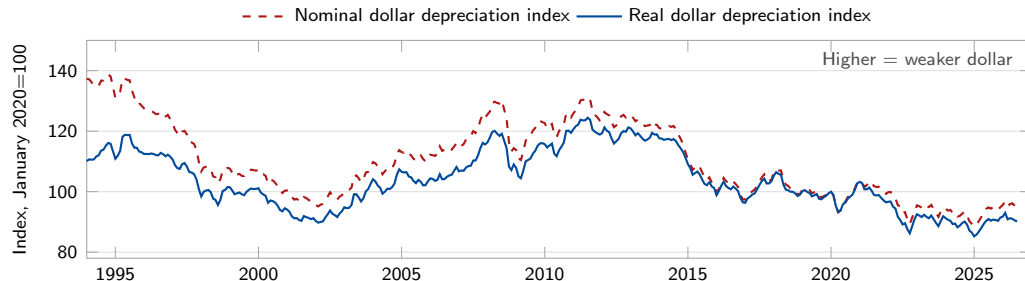
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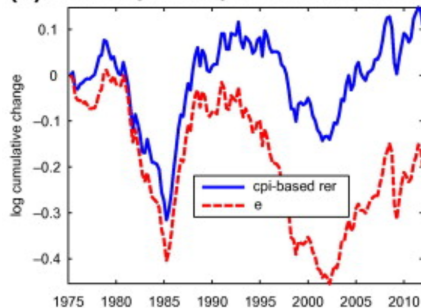
Updated evidence: Broad trade-weighted indices, January 1994–July 2026.



Reading the chart: Dollar swings appear in both series. Relative consumer prices create a slower-moving gap between them. Relative PPP does not hold continuously.

Source: BIS via FRED: NBUSBIS and RBUSBIS, downloaded 14 September 2026. Each series is inverted and rebased to January 2020=100 so that a rise means dollar depreciation, as with our Q . 100 is a reference date, not fair value.

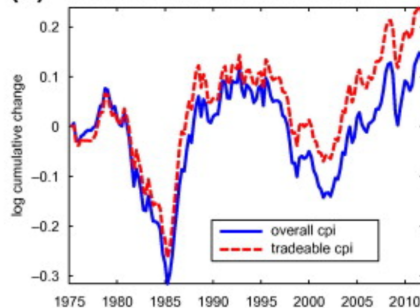
(a) US Trade Weighted Exchange Rate: Nominal and Real CPI-based



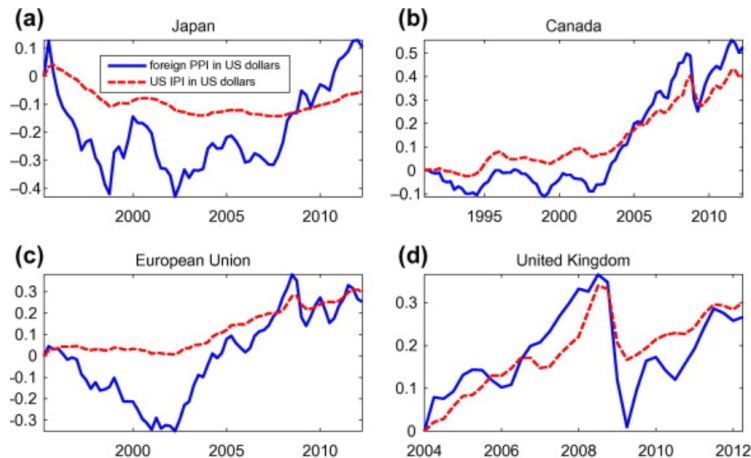
Left: Nominal and CPI-based real dollar movements share large swings. Consumer prices do not immediately offset currency movements.

Source: Burstein & Gopinath (2014), historical sample ending around 2012. Log cumulative changes, dollar depreciation upward.

(b) US CPI-based RER: Overall and tradeable-based



Right: Overall and tradable CPI-based real rates also move together. Nontradable services alone cannot explain all short-run real exchange-rate variation.



Blue: Foreign producer prices expressed in US dollars.

Red: US import prices from that source, also in dollars.

Firms can absorb currency changes in margins, and contracts can delay repricing. For example, dollar depreciation has incomplete pass-through to price paid by a US importer

Source: Burstein & Gopinath (2014). Historical log cumulative changes. PPI and import-price baskets differ.

Horizon	Useful starting point	What could change the forecast?
Days to weeks	News relative to expectations, positioning, funding conditions	A policy surprise or a rapid closing of crowded trades
Months	Relative policy paths, growth, inflation, and external payments	A different inflation or growth path than the one already priced
Years	Relative inflation and the sustainable real exchange rate	Productivity shifts, terms of trade, or persistent demand changes

A usable forecast states its assumptions and alternatives. Compare its performance with a simple unchanged-spot forecast, using only information available at the time.

Under risk neutrality, rational expectations, and ex ante relative PPP, combine the Fisher effect, UIP, and expected inflation:

- Nominal Fisher Effect: $i_H \approx r_H^e + \pi_H^e$ and $i_F \approx r_F^e + \pi_F^e$.
- Uncovered Interest Parity: $\mathbb{E}[\Delta s] \approx i_H - i_F$.
- Relative PPP: $\mathbb{E}[\Delta s] \approx \pi_H^e - \pi_F^e$.

Combining these equations yields Real Interest Parity (RIP):

$$r_H^e = r_F^e$$

These joint assumptions imply approximately equal expected real rates. Equal real rates alone do not establish that all the assumptions hold.

Consider these one-year rates: $i_{US} = 5\%$, $i_{UK} = 3\%$, $\pi_{US}^e = 2\%$, and $\pi_{UK}^e = 1\%$.

$$r_{US}^e \approx 5\% - 2\% = 3\%, \quad r_{UK}^e \approx 3\% - 1\% = 2\%$$

- UIP predicts approximately 2% nominal home depreciation.
- Ex ante relative PPP predicts approximately 1% nominal home depreciation.
- These predictions cannot both hold with the stated rates. If UIP holds, expected real home depreciation is approximately $2\% - 1\% = 1\%$.

For each claim, write “true” or “false” and one sentence explaining your answer.

- ① A currency at a forward premium will appreciate in the spot market.
- ② Buying 1,000 foreign units spot requires selling 1,000 units forward to hedge a deposit earning 2%.
- ③ A credible peg makes all home and anchor-currency interest rates identical.
- ④ A real exchange rate of 0.90 implies that the home currency will depreciate by 10% next year.
- ⑤ Higher factory productivity can raise both the consumption price level and workers' purchasing power.

- ① **False.** The forward rate prices a contract. An expected future spot rate requires additional assumptions.
- ② **False.** Sell 1,020 foreign units forward to cover principal and interest at the deposit's maturity.
- ③ **False.** Rates are linked, but forward points, liquidity, credit terms, and the maturity can support differences.
- ④ **False.** $Q = 0.90$ describes a relative basket price; it does not identify the timing or size of a future nominal movement.
- ⑤ **True.** Higher factory wages can raise service prices while wages rise faster than the cost of the consumption basket.

Condition	Equation	Reason or assumption
Covered Interest Parity (CIP)	$\frac{F - S}{S} \approx i_H - i_F$	No-arbitrage benchmark
Uncovered Interest Parity (UIP)	$\mathbb{E} \left[\frac{\Delta S}{S} \right] \approx i_H - i_F$	Risk neutrality / expectations
Absolute PPP	$S = \frac{P_H}{P_F}$	Frictionless trade / LOP
Relative PPP	$\frac{\Delta S}{S} \approx \pi_H - \pi_F$	Long-run inflation adjustment
Real Interest Parity (RIP)	$r_H^e = r_F^e$	Fisher + UIP + ex ante PPP

Today: We priced hedged cash flows, formed an unhedged exchange-rate view, and connected productivity and goods prices to the real exchange rate.

- **Expectations:** A news release matters through what it changes about the economic outlook. Higher rates can accompany appreciation today and expected depreciation later.
- **Risk:** An attractive expected carry return can coexist with a large currency loss.
- **Valuation:** Relative prices can differ for economic reasons. A higher price level can accompany higher real wages.

Week 3 develops overshooting, tests currency strategies, and studies policy regimes and intervention using these foundations.

Practice: reverse CIA

Practice: order-book depth

Practice: peg and inflation

Fama Regression: Reading a Small Sample

Hypothetical example: Each observation covers the same one-year horizon. Let $x = (F - S)/S$ and $y = (S_T - S)/S$, both expressed in percentage points.

Observation	Forward premium x	Realized spot change y
A	-2	+1
B	0	0
C	+2	-1

- ① Find sample means $\bar{x}$ and $\bar{y}$, and slope $\hat{\beta} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$.
- ② Under CIP, which currency has the higher interest rate in observation C?
- ③ Does the realized exchange-rate movement in C match the UIP prediction?

$$\bar{x} = \bar{y} = 0, \quad \hat{\beta} = \frac{(-2)(1) + (2)(-1)}{(-2)^2 + 2^2} = -0.5, \quad \hat{\alpha} = 0$$

In observation C, $x = +2\%$ implies a higher *home* interest rate under CIP.

UIP predicts home depreciation, but in this example, $y = -1\%$ corresponds to home appreciation.

- A negative slope reverses the predicted relationship in this sample.
- Three observations do not establish a population result. Estimates vary by currency, sample, and horizon.
- Positive average carry returns do not establish a riskless arbitrage.

Source: For variation in actual estimates across periods, see Bussière, Chinn, Ferrara, Heipertz, The New Fama Puzzle (2022 revision).

Appendix: Step-by-Step CIA Calculation

Starting by borrowing US\$1,000,000:

- ① **Borrow Domestic Capital:** Borrow \$1,000,000 USD at 2.0%.
Repayment obligation = \$1,020,000 USD.

- ② **Convert Spot:** Convert USD to GBP at $S_t = 1.3000$:

$$\frac{\$1,000,000}{1.3000} = 769,230.77 \text{ GBP.}$$

- ③ **Lend Abroad:** Lend GBP in UK money market at 1.0%:

$$769,230.77 \times 1.01 = 776,923.08 \text{ GBP.}$$

- ④ **Forward Hedge:** Sell GBP proceeds forward at $F_{t,1} = 1.3200$:

$$776,923.08 \times 1.3200 = \$1,025,538.46 \text{ USD.}$$

- ⑤ **Net Profit:** $\$1,025,538.46 - \$1,020,000 = +\$5,538.46$ USD.

Appendix: Bid-Ask No-Arbitrage Bounds

To prevent profitable covered interest arbitrage in either direction, quotes must satisfy two execution bounds:

Bound 1 (Borrow Home, Lend Foreign):

$$\frac{F_{\text{bid}}}{S_{\text{ask}}} (1 + i_F^{\text{lend}}) \leq 1 + i_H^{\text{borrow}}$$

Bound 2 (Borrow Foreign, Lend Home):

$$\frac{F_{\text{ask}}}{S_{\text{bid}}} (1 + i_F^{\text{borrow}}) \geq 1 + i_H^{\text{lend}}$$

Inside this band, transaction costs eat up any apparent CIP profit margin.

Reserve Exercise: Reverse CIA with Depth

One-year GBP/USD executable quotes (USD per GBP):

	Price	GBP size
Best spot bid / ask	1.300 / 1.302	1,000 each
Best forward bid / ask	1.298 / 1.300	808 each

USD borrowing/lending rates: 3% / 2%. GBP borrowing/lending rates: 1% / 0.5%.

- ① Test both arbitrage directions using the correct funding rates.
- ② If one works, size the trade at the best quotes and calculate USD profit.
- ③ What limits the GBP principal? How does an additional 50 bp GBP borrowing cost change the result?

Reserve Solution: Reverse CIA with Depth

Borrow GBP, lend USD: $1.300(1.02) - 1.300(1.01) = 0.013$ USD per GBP borrowed. The opposite direction loses: $1.298(1.005) - 1.302(1.03) = -0.03657$ USD per GBP deposited.

$$q_{\text{GBP borrowed}} = \min(1,000, 808/1.01) = 800$$

Sell GBP 800 for USD 1,040; the USD deposit grows to 1,060.80. Buy GBP 808 forward for USD 1,050.40 and repay the GBP debt.

$$\text{Profit} = 1,060.80 - 1,050.40 = \text{USD } 10.40$$

With GBP borrowing at 1.5%, re-size to $808/1.015 = 796.0591$ GBP. The margin becomes $1.300(1.02 - 1.015) = 0.0065$ USD per GBP; profit is $796.0591(0.0065) = \text{USD } 5.17$.

Reserve Exercise: A Change in Forward Depth

Return to the AUD/HKD classroom order books: spot asks are 6.000 for 800 AUD, 6.010 for 700, and 6.020 for 1,000. Forward bids remain 6.200, 6.190, and 6.180. Now the best forward bid offers **1,010**

AUD delivery rather than 606. The second level still offers 909 AUD delivery. Rates remain 4% HKD and 1% AUD for one year, with borrowing at deposit rates.

- ① Which leg binds on the first tranche?
- ② Walk the books until the next trade becomes unprofitable.
- ③ How do total spot purchases and total profit compare with the original example?

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Reserve Solution: A Change in Forward Depth

Spot ask	Forward bid	Spot AUD	Forward AUD	HKD profit
6.000	6.200	800	808	17.60
6.010	6.200	200	202	2.32
6.010	6.190	500	505	0.75
Total		1,500	1,515	20.67

The spot leg binds first. At the next prices, $1.01(6.190) - 1.04(6.020) = -0.0089$ HKD per AUD. Stop. Spot volume stays at 1,500 AUD, but profit rises by $20.67 - 16.63 = \text{HKD } 4.04$. An extra 404 AUD of forward delivery receives 6.200 rather than 6.190. **Interpretation:** Greater depth can improve the execution price without changing the optimal total quantity.

Reserve Exercise: Nominal Stability and Real Appreciation

Home is Hong Kong; foreign is the US. The same basket costs HKD 780 in Hong Kong and USD 100 in the US at $S_0 = 7.80$ HKD per USD. Over the next year, Hong Kong basket inflation is 5%, US basket inflation is 2%, and the nominal rate stays at 7.80.

- ① Compute Q_0 and Q_1 . Which currency appreciates in real terms?
- ② What nominal exchange rate would keep Q constant?
- ③ Explain why a fixed nominal exchange rate does not guarantee stable international purchasing power.

Reserve Solution: Nominal Stability and Real Appreciation

$$Q_0 = \frac{7.8(100)}{780} = 1, \quad Q_1 = \frac{7.8(102)}{819} = 0.971429$$

The HKD appreciates in real terms: Q falls by 2.86%. Hong Kong's basket becomes more expensive relative to the US basket.

$$S_1^{\text{constant } Q} = \frac{819}{102} = 8.029412 \text{ HKD/USD}$$

- Constant Q would require the USD price in HKD to rise 2.94%.
- The actual nominal rate is fixed in this scenario; the inflation gap instead changes the real exchange rate.
- A nominal peg disciplines the exchange rate, while domestic prices can still move relative to foreign prices.